

深 圳 市 金 逸 晨 电 子 有 限 公 司

LCD MODULE

MODULE NO. :

GMT020-02-8P

Customer:

Approved By(核准) :

深圳市金逸晨电子有限公司

Approved By(核准) :

Checked By(审核) :

Prepared By(编写) :

(修订记录)

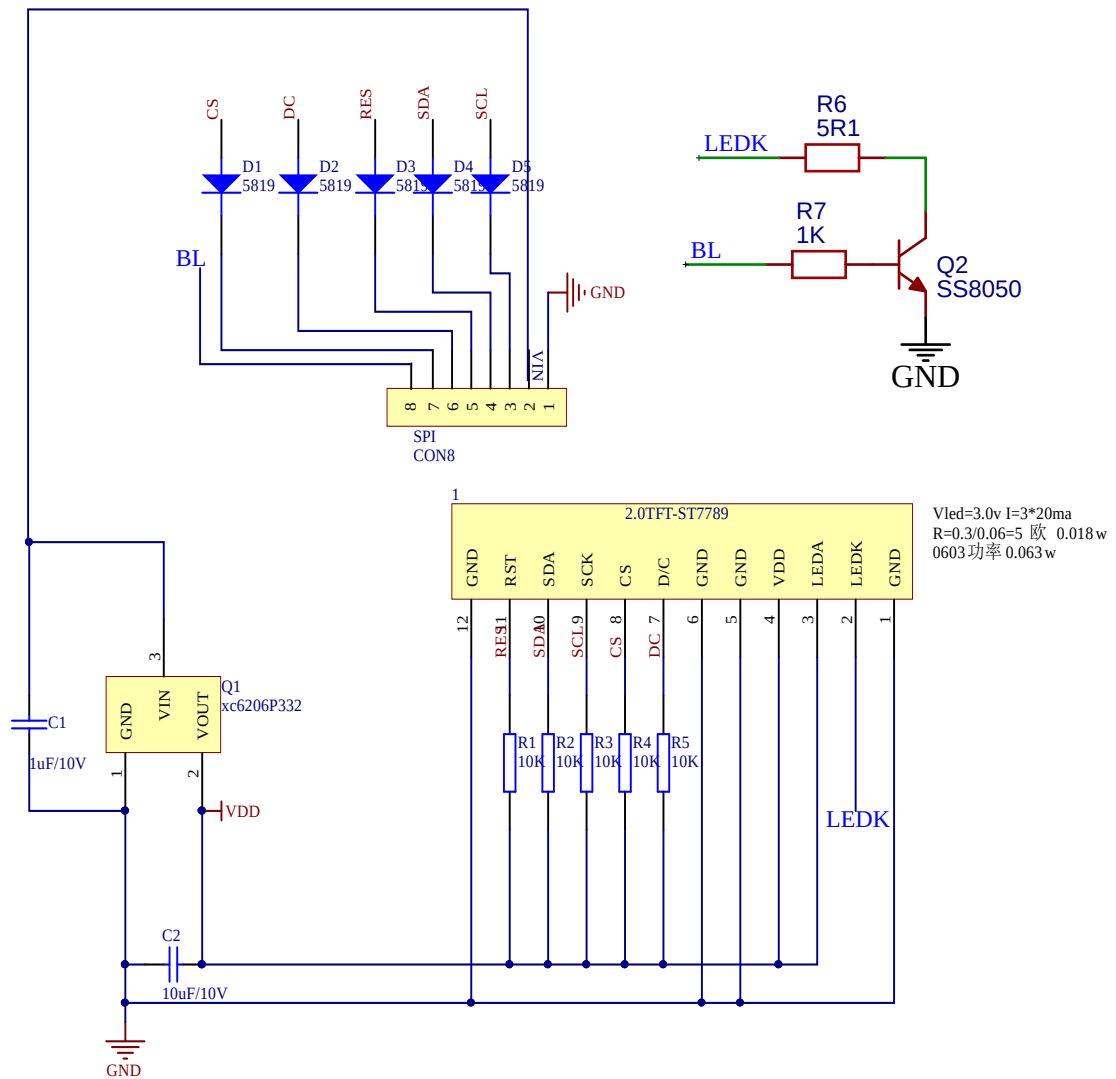
Part Number (产品型号)	Revision (版本)	Revision Content (修订内容)	Revised on (修订日期)
GMT020-02-8P	V1.0	第一次发行 (PCB:V1.21)	2024-11-5



CONTENTS

1. 外形图	4
2. 功能&特性	4
3. 机械规格	4
4. 原理图	5
5. 引脚说明	5
6. 电气特性	6
6.1 DC电气特性	6
6.2 AC电气特性	7
7. 指令表	8

4. 原理图:



注意：电路及元件值仅作参考

5. 引脚说明:

Pin No.	Symbol	Function
1	GND	电源负
2	VCC	电源正
3	SCL	时钟
4	SDA	数据输入
5	RST	复位
6	DC	命令/数据
7	CS	片选
8	BL	背光控制

6. 电气特性

6-1 DC 电气特性

6-1.1. ABSOLUTE MAXIMUM RATINGS

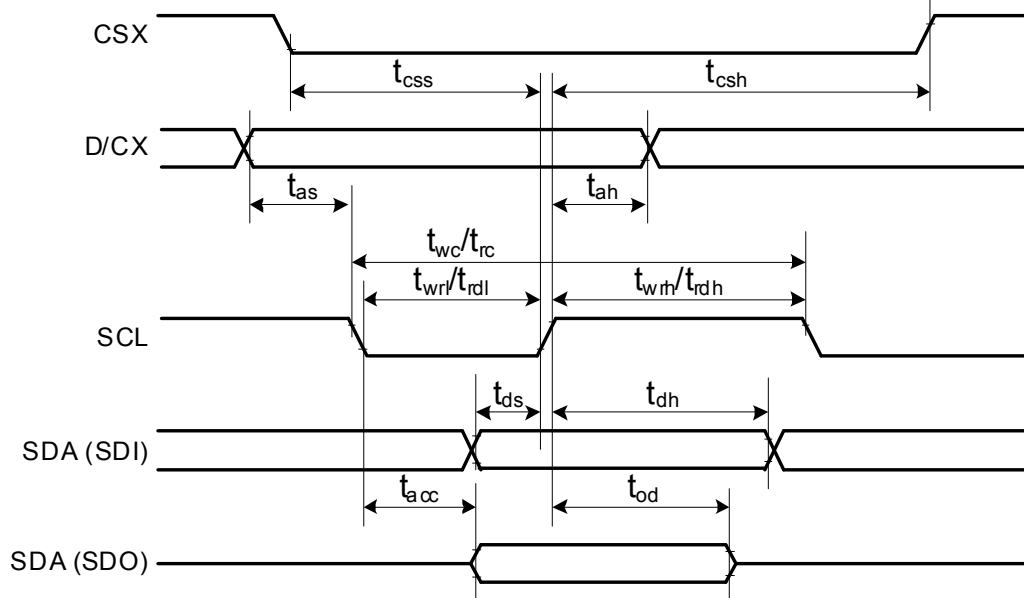
Parameter	Symbol	Min	Max	Unit
Supply voltage for logic	V_{DD}	-0.3	3.0	V
Input voltage for logic	V_{IN}	-0.5	$V_{DD} + 0.3$	V
Supply current (One LED)	I_{LED}		20	mA
Operating temperature	T_{OP}	-20	+70	°C
Storage temperature	T_{ST}	-30	+80	°C

6-1.2. ELECTRICAL CHARACTERISTICS

Item	Symbol	Min	Typ	Max	Unit	Applicable terminal
Supply voltage for logic	V_{DD}	2.6	2.8	3.0	V	V_{DD}
Input voltage	V_{IL}	-0.3	-	$0.2 V_{DD}$	V	
	V_{IH}	$0.8 V_{DD}$	-	V_{DD}	V	
Input leakage current	I_{LKG}				μA	
LED Forward voltage	V_f	2.9	3.0	3.1	V	--
Input backlight current	I_{LED}	-	60	--	mA	With 4x LED

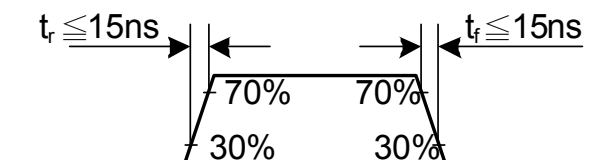
6-2 AC电气特性

Display Serial Interface Timing Characteristics (4-line SPI system)



Signal	Symbol	Parameter	min	max	Unit	Description
CSX	t_{css}	Chip select time (Write)	40	-	ns	
	t_{csh}	Chip select hold time (Read)	40	-	ns	
SCL	t_{wc}	Serial clock cycle (Write)	100	-	ns	
	t_{wrh}	SCL "H" pulse width (Write)	40	-	ns	
	t_{wrl}	SCL "L" pulse width (Write)	40	-	ns	
	t_{rc}	Serial clock cycle (Read)	150	-	ns	
	t_{rdh}	SCL "H" pulse width (Read)	60	-	ns	
	t_{rdl}	SCL "L" pulse width (Read)	60	-	ns	
D/CX	t_{as}	D/CX setup time	10	-		
	t_{ah}	D/CX hold time (Write / Read)	10	-		
SDA / SDI (Input)	t_{ds}	Data setup time (Write)	30	-	ns	
	t_{dh}	Data hold time (Write)	30	-	ns	
SDA / SDO (Output)	t_{acc}	Access time (Read)	10	-	ns	For maximum CL=30pF
	t_{od}	Output disable time (Read)	10	50	ns	For minimum CL=8pF

Note: $T_a = 25\text{ }^{\circ}\text{C}$, $V_{DDI}=1.65\text{V to }3.3\text{V}$, $V_{CI}=2.5\text{V to }3.3\text{V}$, $AGND=VSS=0\text{V}$



7. 指令表 COMMAND TABLE

参阅ST7789V2（IC资料）：

SYSTEM FUNCTION COMMAND TABLE 1 (Page158~Page254)

SYSTEM FUNCTION COMMAND TABLE 1 (Page255~Page314)